

IOT-BASED SMART IRRIGATION SYSTEM FOR URBAN FARMING

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ABSTRACT

This study develops an IoT-based automated irrigation system using Arduino, soil moisture sensors, and a mobile app, reducing water consumption by 34% while maintaining comparable crop yield.

INTRODUCTION

Urban farming in the Philippines faces water management challenges due to irregular rainfall and reliance on manual irrigation. Over-irrigation wastes resources; under-irrigation reduces yield. IoT-enabled precision irrigation offers a data-driven solution. This study designs and evaluates an automated irrigation system for a 200 sqm rooftop garden at Imus City Hall, targeting water efficiency improvement.

METHODOLOGY

The system comprised Arduino Mega 2560 microcontroller, capacitive soil moisture sensors (4 units), a 12V solenoid valve, DHT22 temperature-humidity sensor, and a 4G GSM module. A threshold-based irrigation algorithm activates watering when soil moisture drops below 35%. A Flutter mobile app provides real-time sensor monitoring and manual override. Data was logged over 90 days (June–August 2023). Evaluation: daily water usage (liters), soil moisture stability (variance), and crop yield (kg/sqm).

RESULTS

The IoT system reduced average daily water consumption from 85L to 56L (34% reduction). Soil moisture variance decreased from $\pm 12\%$ to $\pm 4\%$, indicating stable crop hydration. Crop yield was 2.1 kg/sqm vs. 2.0 kg/sqm for the manual control plot (not statistically significant, $p=0.31$). System uptime was 97.2% over the 90-day evaluation period.

DISCUSSION

The 34% water reduction demonstrates meaningful efficiency gains without yield compromise. The non-significant yield difference suggests that threshold-based irrigation is comparable to experienced manual irrigation for common crops. Future work should integrate weather API forecasting to pre-emptively adjust thresholds and evaluate multi-crop scenarios.